```c
#include <stdio.h>

int main() {
    int page_table[4] = {1, -1, 3, -1};
    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    } else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

```
yashwanth@Yashwanth:~$ gcc pagetable.c -o pt
yashwanth@Yashwanth:~$ ./pt
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
yashwanth@Yashwanth:~$
```

```
 yashwanth@Yashwanth:~$ cat nullpointerdere.c
#include <stdio.h>

int main() {
    int *p = NULL;      // p points to nothing
    printf("Before accessing memory\n");
    *p = 100;           // ❌ Attempt to write to NULL → crash
    printf("After accessing memory\n");
    return 0;
}
```

```
yashwanth@Yashwanth:~$ gcc nullpointerdere.c -o np
yashwanth@Yashwanth:~$ ./np
Before accessing memory
Segmentation fault (core dumped)
```

```c
#include <stdio.h>

int main() {
    int page_table[] = {3, 5, 1, 7}; // Example page table
    int page_size = 100;             // Size of each page/frame

    int virtual_address = 250;       // Example virtual address

    // Split virtual address into page number and offset
    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    // Translate page number to frame number using page table
    int frame = page_table[page];

    // Calculate physical address
    int physical_address = frame * page_size + offset;

    // Display results
    printf("Virtual Address  = %d\n", virtual_address);
    printf("Page Number      = %d\n", page);
    printf("Offset           = %d\n", offset);
    printf("Frame Number     = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}
```

```
yashwanth@Yashwanth:~$ gcc addresstrans.c -o at
yashwanth@Yashwanth:~$ ./at
Virtual Address  = 250
Page Number      = 2
Offset           = 50
Frame Number     = 1
Physical Address = 150
```